

Benchmarking of Phase Locked Loop based Synchronization Techniques for Grid-Connected Inverter Systems

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Abstract—Grid-connected renewables are increasingly developed in recent years, e.g. wind turbine systems and photovoltaic systems. Synchronization of the injected current with the grid is mandatory. However, grid disturbances like voltage sags, harmonics, and frequency deviations may occur during operation, becoming inevitable challenges to the synchronization of the grid-connected renewable energy systems. In order to ensure the quality of the power generation from the renewables, robust and reliable synchronization methods are in demand. Among the prior-art solutions, Phase Locked Loop (PLL) based synchronization methods have gained much popularity in grid-connected applications. However, an appropriate selection and thus a proper design of the selected PLL synchronization remain of interest in practice, especially for single-phase systems. Therefore, in this paper, a benchmarking of the main PLL synchronization methods for single-phase grid-connected inverter systems in terms of accuracy, dynamic response, harmonic immunity, etc., has been conducted. Experiments on a 1-kW single-phase grid-connected system, suffering from different grid disturbances, are performed for the benchmarking. The experimental results have verified the discussions.

Index Terms—Phase locked loop (PLL), synchronization, T/4 delay PLL, inverse park transform PLL, enhanced PLL (EPLL), second order generalized integrator PLL (SOGI-PLL), multi-harmonic decoupling cell PLL.

I. INTRODUCTION

Power electronics technology has enabled an increasing integration of renewable energy into the grid [1]–[3], e.g. wind turbine and photovoltaic systems. However, due to the intermittency of renewable energy, it has also brought harmonic challenges to the grid [4]–[6], as a fluctuating power is continuously injected to the grid. Since the inner current controller of a typical two-cascaded control system [1], [4] is responsible for shaping the current (e.g., power quality issues) in such applications, efforts have to be devoted to the control of the feed-in grid current, which has to be synchronized with the grid voltage using a synchronization system. Fig. 1 demonstrates the significance of synchronization in the entire control of a single-phase system. As it can be seen in Fig. 1, the information provided by a synchronization is of ultimate

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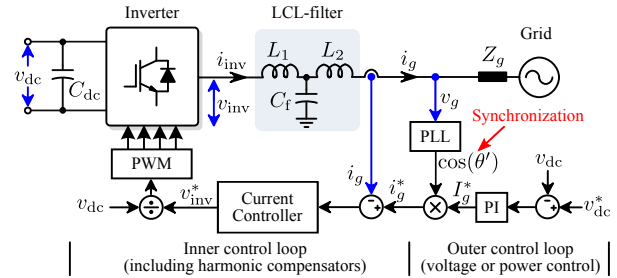


Fig. 1. Overall control structure (dual-loop) of a single-phase grid-connected inverter system with an LCL-filter.

importance, and it is used at different levels of the entire control system, e.g. for a reference frame transformation (the $\alpha\beta$ -stationary frame $\rightarrow$ the dq -rotating frame).

In the literature, a vast of synchronization methods have been reported [4], [7]–[17], [29]. These prior-art synchronization methods can be categorized into a) mathematical analysis methods (e.g., grid synchronization based on the Discrete Fourier Transform - DFT) [16], [18]–[20] and b) Phase Locked Loop (PLL) based synchronization techniques. The first category are based on signal processing techniques, e.g. the DFT and Hilbert transform analysis, which are commonly implemented in a digital controller, and thus has a strict requirement of the sampling rate [4], [8], [21]. In contrast, a PLL synchronization method is a closed-loop system, which is widely used in the grid-connected systems [4], [12]. However, for single-phase applications, where only the grid voltage v_g is measured, it requires more dedications to the synchronization.

Advanced control strategies can thus be applied to an inverter system, while its performance significantly relies on the dynamics of the synchronization, as shown in Fig. 1. Both the mathematical analysis and the PLL based synchronization are able to provide accurate and fast information for the control in the case of a normal grid condition [8], [14], [18]. However, the grid voltage cannot always be maintained as “constant” in terms of amplitude, frequency, and phase, due to multiple eventualities like continuous connection and disconnection of loads and fault to ground [22], [23]. Together with the harmonics in the grid voltage (e.g., due to non-linear loads), a big

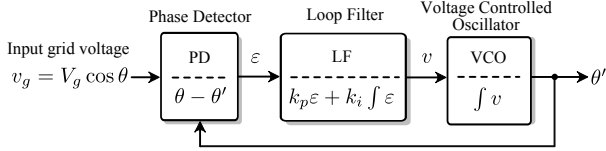


Fig. 2. Basic block diagram of a phase locked loop system for grid-connected inverter systems, where V_g is the grid voltage amplitude.

obstacle for synchronization, especially for the Fourier based method [16], [24], has been imposed on. Thus, it calls for an advancement of these synchronization techniques in order to enhance the entire system performance, requiring a clear identification of the pros and cons of the synchronization methods. Alternatively, a benchmarking of the most commonly-employed PLL techniques could contribute to not only an enhancement of the PLLs but also a development of new PLLs. As a result, an advanced synchronization system will ensure a more reliable control of the injected current and thus a more reliable and stable operation of the entire grid-connected inverter system.

In light of the above issues, a benchmarking of several selected PLL based synchronization methods is conducted in this paper. Firstly, the basics of the single-phase PLL system are presented in § II, including the small signal modelling and basic design considerations. Then, § III gives a description of the selected PLL synchronization methods, including the $T/4$ Delay PLL, the Inverse Park Transform PLL (IPT-PLL), the Enhanced PLL (EPLL), the Second Order Generalized Integrator based PLL (SOGI-PLL), and the Multi-Harmonic Decoupling Cell based PLL (MHDC-PLL). Followingly, those synchronization techniques are benchmarked in terms of accuracy, dynamic response, harmonic immunity, and etc. by experimental tests in § IV, where the grid suffers from various disturbances. A conclusion is then drawn.

II. BASICS OF SINGLE-PHASE PLL SYSTEMS

As it can be seen in Fig. 1, the PLL system plays a key role in the entire control loop of grid-connected inverter systems. Typically, a PLL system consists of a Phase Detector (PD), a Loop Filter (LF) which is performed by a Proportional Integrator (PI) controller, and a Voltage Controlled Oscillator (VCO). Fig. 2 represents the basic block diagrams of a PLL system that is widely utilized in grid-connected applications. Accordingly, the transfer function of the PLL system $G_{PLL}(s)$ can be described as,

$$G_{PLL}(s) = \frac{\theta'(s)}{\theta(s)} = \frac{k_p s + k_i}{s^2 + k_p s + k_i} \quad (1)$$

where θ is the grid voltage phase, θ' is the locked grid voltage phase, k_p and k_i are the proportional and integral gains of the PI controller (i.e., the LF).

It can be seen from (1) that the PLL system is a typical second order system [4]. Therefore, the damping ratio ξ and the undamped natural frequency ω_n of (1) can be calculated by,

$$\xi = \frac{k_p}{2\omega_n}, \text{ and } \omega_n = \sqrt{k_i}.$$

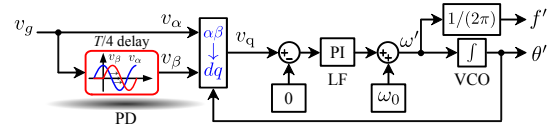


Fig. 3. Detailed structure of a phase locked loop system by introducing a quarter phase delay ($T/4$ Delay PLL), where $\omega_0 = 2\pi f_0$ with f_0 being the nominal grid frequency.

Subsequently, the settling time of the PLL system can be approximated as,

$$t_s = \frac{4.6}{\xi\omega_n} = \frac{9.2}{k_p} \quad (2)$$

which can be used to benchmark the transient performance of different PLL systems, and also can be taken as guidelines for tuning the LF parameters:

$$k_p = \frac{9.2}{t_s}, \text{ and } k_i = \left(\frac{k_p}{2\xi}\right)^2 \quad (3)$$

where $\xi = \sqrt{2}/2$ is typically chosen for a satisfactory and optimal damping.

In the case that the control parameters of the LF (i.e., k_p and k_i) are set to be identical in different PLLs, the performance of the PLL synchronization methods will strongly rely on the configurations of the PD system. A sinusoidal multiplier is the most intuitive way to the implementation of a PD system, but it introduces double-frequency harmonics in the closed-loop system [13], which cannot be fully eliminated by the LF (i.e., the PI controller). Therefore, the task of advancing a PLL system is shifted to improve the detection of the phase error (i.e., $\varepsilon = \theta' - \theta$) using the input grid voltage v_g and feedback signals (e.g., the estimated phase θ'). The following demonstrates how the PD systems are constructed in the most commonly employed single-phase PLL techniques.

III. SELECTED PLL SYNCHRONIZATION METHODS

As previously discussed, a number of single-phase PLL based synchronization methods have been developed in the literature. In this section, the most popular PLL systems are presented, including their basic design guidelines.

A. $T/4$ Delay PLL

An alternative phase detection can be achieved with the aid of the Park transform ($\alpha\beta \rightarrow dq$), where a virtual system (β variable) that is in-quadrature with the grid voltage v_g (α variable) is required in single-phase applications. Simply, a quarter delay of the input grid voltage is a possibility, being the $T/4$ Delay PLL, as it is shown in Fig. 3, where T is the known fundamental period of the input grid voltage v_g .

Specifically, on an assumption that the grid voltage is purely sinusoidal, i.e., $v_g(t) = V_g \cos(\theta) = V_g \cos(\omega t + \varphi_0)$ with ω being the grid angular frequency and φ_0 being the initial phase angle, applying the Park transform yields,

$$\mathbf{v}_{dq} = \overbrace{\begin{bmatrix} \cos(\theta') & \sin(\theta') \\ -\sin(\theta') & \cos(\theta') \end{bmatrix}}^{T_p} \mathbf{v}_{\alpha\beta} \approx V_g \begin{bmatrix} 1 \\ \varepsilon \end{bmatrix} \quad (4)$$

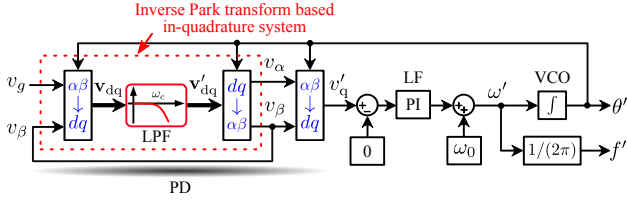


Fig. 4. Block diagram of the inverse park transform based phase locked loop (IPT-PLL).

where T_p is the Park transform matrix. Eq. (4) shows that the detected phase error ε (i.e., v_q/V_g) can be regulated by a PI controller so that the grid voltage phase θ is locked as θ' in the steady-state. In addition, the grid voltage amplitude $V_g = v_d$ and frequency $f' = \omega'/(2\pi)$ can also be obtained according to (4) and Fig. 3, respectively.

In regards to the implementation of the $T/4$ Delay PLL, a fixed delay of a quarter period (i.e., $T/4 = 1/(4f_0)$ with f_0 being the nominal grid frequency) is normally adopted for simplicity. As a result, the dependence of the grid voltage frequency f_0 to create the virtual in-quadrature system makes the $T/4$ Delay PLL not very suitable for the single-phase applications, where the grid voltage is subjected to frequency variations [25]. Moreover, the background distortions will directly propagate to the LF when the input voltage is delayed for a quarter period. This becomes another big challenge to the $T/4$ Delay PLL.

B. Inverse Park Transform PLL (IPT-PLL)

Another possibility to detect the phase error seems to be a good one for single-phase systems, and it is based on the Inverse Park Transform (IPT, $dq \rightarrow \alpha\beta$) [4]. Fig. 4 shows the block diagram of an IPT based PLL system (IPT-PLL). When compared to the $T/4$ Delay PLL, the IPT-PLL requires two additional Low Pass Filters (LPF), and thus certain harmonics in the grid voltage will not propagate to the LF, contributing to a good harmonic rejection. Thus, from the harmonic immunity point of view, the IPT-PLL is better than the $T/4$ Delay PLL.

According to Fig. 4, the PD structure of the IPT-PLL system can be described by the following:

$$\mathbf{v}_{dq}(s) = \mathbf{T}_p(s) \begin{bmatrix} v_g(s) \\ v_\beta(s) \end{bmatrix}, \mathbf{v}'_{dq}(s) = \mathbf{T}_p(s) \begin{bmatrix} v_\alpha(s) \\ v_\beta(s) \end{bmatrix} \quad (5)$$

and

$$\mathbf{v}'_{dq}(s) = G_{LPF}(s) \mathbf{v}_{dq}(s) = \frac{\omega_c}{s + \omega_c} \mathbf{v}_{dq}(s) \quad (6)$$

where $T_p(s)$ is the Laplace form of T_p shown in (4), and $G_{LPF}(s)$ is the transfer function of the first-order LPF with ω_c being the corresponding cut-off angular frequency. Then, exploiting the Euler formula and the Laplace property for the frequency shifting yields [4],

$$\mathbf{v}_{\alpha\beta}(s) = \begin{bmatrix} v_\alpha(s) \\ v_\beta(s) \end{bmatrix} = \begin{bmatrix} \frac{\omega_c s}{s^2 + \omega_c s + \omega'^2} \\ \frac{\omega_c \omega'}{s^2 + \omega_c s + \omega'^2} \end{bmatrix} v_g(s) \quad (7)$$

which indicates that the performance of the IPT-PLL system is highly dependent on the LPF, $G_{LPF}(s)$.

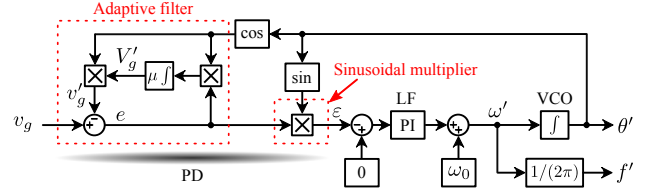


Fig. 5. Enhanced phase locked loop (EPLL) system based on an adaptive filtering technique [9].

A design parameter of the IPT-PLL system can then be defined as $k_{ipt} = \omega_c/\omega'$. In accordance to (7), $v_g(s)$ -to- $v_\alpha(s)$ and $v_g(s)$ -to- $v_\beta(s)$ represent a second-order band-pass and low-pass filter, respectively. Consequently, the design parameter k_{ipt} should be equal to $\sqrt{2}$ in order to ensure an optimal damping of the second-order filters in terms of good settling time and overshooting in the dynamics, and accordingly the cut-off frequency ω_c of the LPF can be set. However, the design parameter k_{ipt} , may need to be tuned slightly in practice when considering the entire control system (e.g., computation delay effects). Nevertheless, the IPT PD system can ensure not only a proper deriving of the in-quadrature system ($\mathbf{v}_{\alpha\beta}$) but also a filtering of the high-order harmonics of the grid voltage (i.e., since $v_g(s)$ -to- $v_\beta(s)$ behaves as a second-order LPF), resulting in a good harmonic immunity, which is in coincidence with the previous discussion.

C. Enhanced PLL (EPLL)

The basic ideas of the above two PLLs fall into the establishment of in-quadrature systems, thus enabling the Park transform to detect the phase error. Using adaptive filtering techniques is another way to the phase detection, as an adaptive filter is able to adjust the parameters automatically according to the error signal and a reference (e.g., $\cos(\theta')$) [4], [9]–[11]. The Enhanced PLL (EPLL), which was introduced in [9], is a typical representative of adaptive filtering based PLL systems. Similar principle has also been implemented in the control of the instantaneous power of a single-phase system [10].

Actually, the EPLL phase detection is achieved using an Adaptive Filter (AF) and a simple sinusoidal multiplier, as it is shown in Fig. 5, so that the EPLL can enhance the performance in contrast to a sinusoidal multiplier based PLL [8]. More specifically, the AF is used to estimate the input voltage v_g according to the detected phase error ε and the locked phase θ' (in order to generate the filter reference $\cos(\theta')$) by minimizing an objective function, e.g., $(v_g - v'_g)^2/2$. After a period, the frequency and phase of the EPLL will be free of oscillations [4].

As it can be observed in Fig. 5, the most important feature of an EPLL is that both the grid voltage amplitude V'_g and the phase θ' of the input voltage v_g can be locked. According to Fig. 5, the estimated grid voltage amplitude V'_g can be expressed as,

$$\dot{V}'_g = \mu e \cos(\theta') \quad (8)$$

in which μ is the control parameter and $e = (v_g - v'_g)$. It is also implied in (8) that the dynamic response of the EPLL

TABLE I
PARAMETERS OF THE SINGLE-PHASE GRID-CONNECTED SYSTEM.

Parameter	Symbol	Value
Rated power	P_n^*	1 kW
DC-link voltage	V_{dc}	450 V
Grid voltage amplitude	V_g	$230 \times \sqrt{2}$ V
Grid nominal frequency	ω_0	$2\pi \times 50$ rad/s
DC-link capacitor	C_{dc}	1100 μ F
LC filter	L_1	3.6 mH
	C_f	2.35 μ F
Transformer leakage inductance	L_g	4 mH
Sampling frequency	f_s	10 kHz
Switching frequency	f_{sw}	10 kHz

TABLE II
CURRENT CONTROLLER AND LOOP FILTER PARAMETERS.

Controller	Symbol	Value
PLL loop filter (PI)	k_p	0.283
	k_i	5.663
PR controller	k_{pr}	22
	k_r	1300
HC controller	$k_r^{3,5}$	1000
	k_r^{7}	600

TABLE III
PARAMETERS OF THE PHASE DETECTOR SYSTEMS OF THE
SELECTED SINGLE-PHASE PLLS.

PLL	Symbol	Value
IPT-PLL	k_{ipt}	1.4
EPLL	τ	8 ms
SOGI-PLL	k	1.4
MHDC-PLL	ω_c	$2\pi \times 50 \times \sqrt{2}$ rad/s
	ω_f	$2\pi \times 50/3$ rad/s

* LPF cut-off freq. in the IPT in-quadrature system.

$THD_{v_g} \approx 0.71\%$), and the experimental results are presented in Fig. 9, where the frequency error $\Delta f = f' - 50$ and v_d error $\Delta v_d = v_d - 1$ (or $v'_d - 1$). Since the capacitor voltage is measured as the grid voltage v_g for synchronization which thus contains switching frequency harmonics, it is observed in Fig. 9 that the outputs of both the $T/4$ Delay PLL and the EPLL consist of high-order harmonics. This indicates the poor harmonic immunity of the $T/4$ Delay PLL and the EPLL systems. Moreover, the $T/4$ Delay PLL is more sensitive to the harmonics, as its in-quadrature system is based on a quarter phase delay of the input voltage, thus inevitably inheriting the harmonics. As a result, in the case of grid-connected applications using the $T/4$ Delay PLL or the EPLL, a LPF may be required and should be incorporated at point "A" shown in Fig. 8(b), which however can affect the dynamics of the entire system. In contrast, the IPT-PLL, the SOGI-PLL, and the MHDC-PLL are all good at eliminating these switching harmonics due to the presence of the low pass filters or the adaptive notch filter.

However, in the case of a very weak grid which contains not only low-order but also high-order harmonics, the PLL systems will be challenged. Fig. 10 further benchmarks the dynamic performances of the selected PLL methods

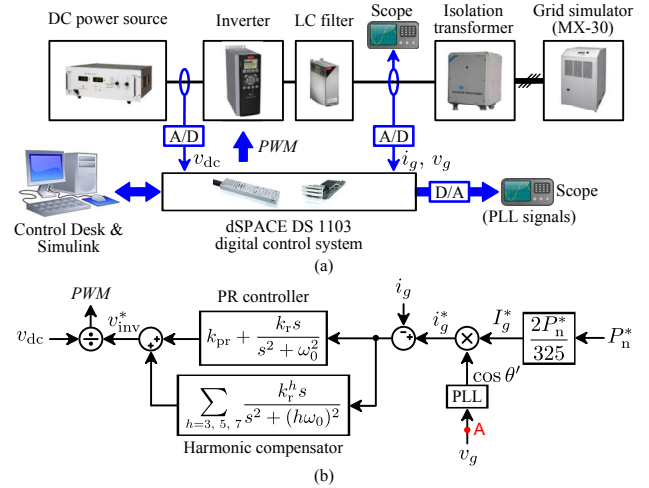


Fig. 8. Experimental setup of a 1-kW single-phase grid-connected system: (a) dSPACE platform and (b) current control system.

when the grid voltage experiences several disturbances. The harmonic sensitivity of the $T/4$ Delay PLL and the EPLL is clearly verified by the results shown in Fig. 10(a), where it also shows that the IPT-PLL and the SOGI-PLL are slightly affected by the low order harmonics of the grid voltage. In contrast, it can be observed in Fig. 10(a) that the estimated output frequency of the MHDC-PLL system is free of oscillations after a short period of transient. This means that the MHDC-PLL is significantly immune to the harmonics in the grid voltage.

Moreover, it can be observed in Figs. 9 and 10 that the performances of the IPT-PLL and the SOGI-PLL systems are quite alike, since the in-quadrature systems of both PLLs have similar filtering capability (i.e., $v_{\alpha\beta}$ -to- v_g) according to (7) and (10). In addition, as it is shown in Fig. 10 (c), the $T/4$ Delay PLL and the MHDC-PLL systems present poor synchronization performances in the case of grid frequency variations, which may occur especially in the micro grid systems due to an injection of a large amount of fluctuating power, e.g., PV and/or wind power. This poor frequency adaptability is because of the adoption of the delay unit with a constant duration for the in-quadrature systems. Nevertheless, the two PLLs present fast dynamics. For the MHDC-PLL, the dynamics are even comparable with the IPT-PLL and the SOGI-PLL in terms of a fast response and a small overshooting, as it is verified by Fig. 10(b) and (d). In all, the test results are in close agreement with the discussions in § III.

In addition to the above verification, a comparison between the SOGI-PLL and the MHDC-PLL has also been conducted with a programmed background distortion of the grid voltage (i.e., $THD_{v_g} = 2.91\%$). In this case study, the HC is disabled in order to compare how the PLL synchronization will impact the injected grid current quality. The experimental results are demonstrated in Fig. 11. It can be seen in Fig. 11 that, if the MHDC-PLL system is adopted as the synchronization, the grid current THD_{i_g} is slightly reduced in contrast to the case when the SOGI-PLL is used. Actually, when looking at the

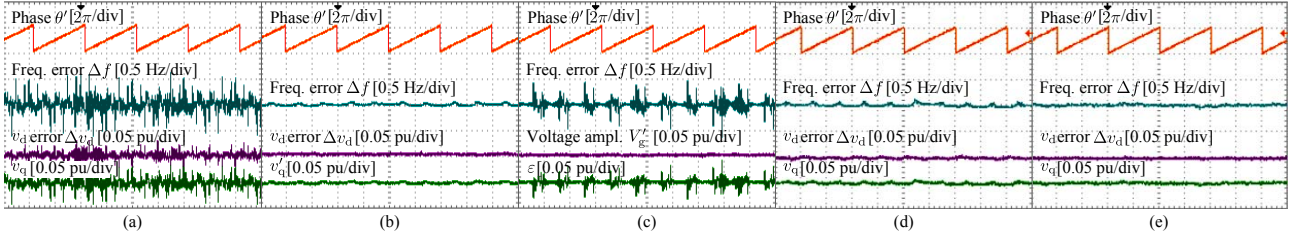


Fig. 9. Steady-state performance of the selected PLL synchronization methods under a nominal grid condition (time [10 ms/div]): (a) $T/4$ Delay PLL, (b) IPT-PLL, (c) EPLL, (d) SOGI-PLL, and (e) MHDC-PLL.

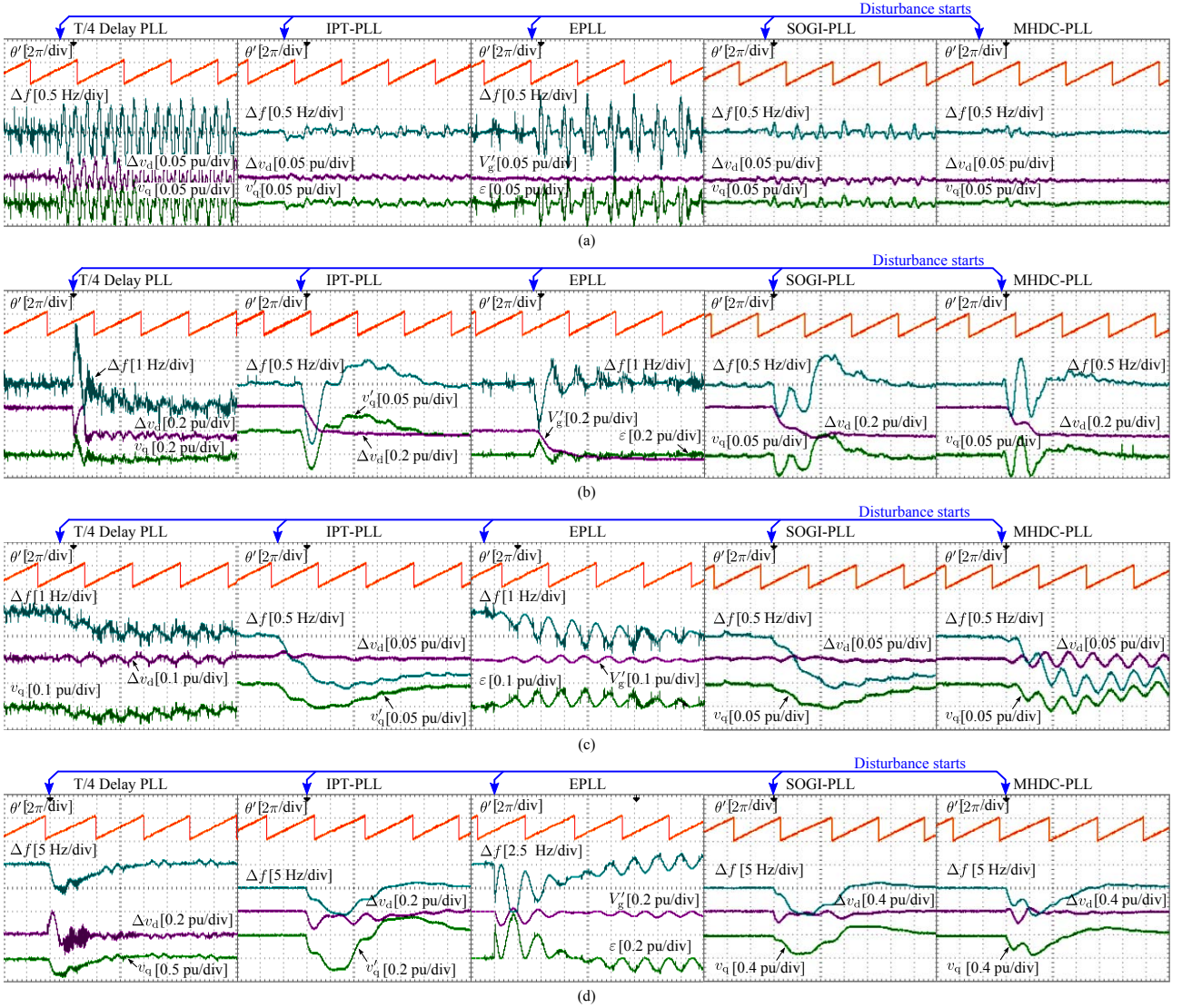


Fig. 10. Dynamic responses of the selected PLL synchronization methods under various grid disturbances (time [10 ms/div]): (a) harmonics (THD_{v_g} changes from 0.71% to 2.91%), (b) voltage sag ($V_g = 0.75$ pu), (c) frequency jump (-0.8 Hz), and (d) phase shift (-30°).

individual low-order harmonics, the same conclusion can be reached. For example, the Root Mean Square (RMS) value of the 7th harmonic is 112 mA when the grid current is synchronized through the SOGI-PLL system, while the 7th harmonic of 106 mA is achieved using the MHDC-PLL system. The experimental results have demonstrated that the synchronization will affect the entire control systems as mentioned in § I. It should be emphasized that efforts can be devoted to the advancement of synchronization

methods in order to achieve a better current injection from the grid-connected inverter systems.

V. CONCLUSION

In this paper, a benchmarking of the most popular PLL methods (i.e., $T/4$ Delay PLL, EPLL, IPT-PLL, SOGI-PLL, and MHDC-PLL) for single-phase grid-connected systems has been presented. Benchmarks include the accuracy, the transient response, the harmonic immunity under

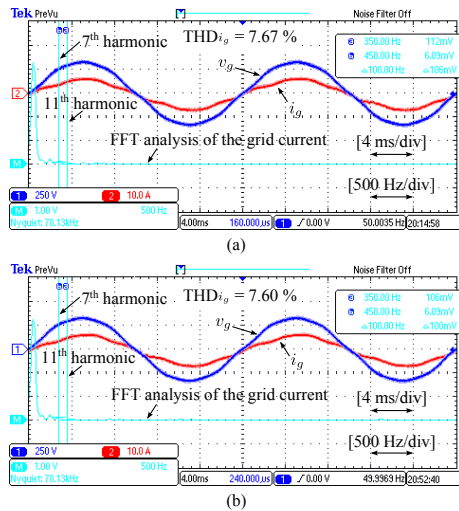


Fig. 11. Experimental results of a 1-kW single-phase grid-connected system with different PLL systems (CH1 - grid voltage v_g [250 V/div], CH2 - grid current i_g [10 A/div], CH M - FFT analysis of the grid current [1 A/div]): (a) SOGI-PLL system and (b) MHDC-PLL.

grid disturbances, and the implementation complexity. As a result, the benchmarking results provide a flexibility to choose an appropriate PLL-based synchronization technique according to a specific application. For example, the SOGI-PLL is suitable for fault ride-through operations in terms of high accuracy and a fast response speed. In contrast, the MHDC-PLL also with fast dynamic responses is good for use in a weak grid that is distorted by non-linear loads, while its performance is poor in response to grid frequency variations due to the T/4 delay mechanism. Experimental tests on a single-phase grid-connected system have supportively verified the benchmarking.

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